

## S05 SOUND REDUCING SURFACES | O (MAX : 2 PT )

**Intent :** Design spaces with sound reducing surfaces to minimize the buildup of speech or other unwanted sound.

**Summary :** This WELL feature requires the use of acoustic materials that absorb and/or block sound to support concentration and reduce reverberation.

**Issue :** Increases in reverberation are commonly caused by lack of acoustical absorption.<sup>1</sup> When spaces are designed with hard reflective ceilings and walls, perception of noise increases with an increase in ambient, reflected noise that decays in level at a longer rate.<sup>2,3</sup> When open collaborative spaces lack acoustical absorption at ceilings or partial height barriers, workers can become distracted by reflected sound, especially if that speech is perceptible.<sup>3-6</sup> Conversely, in learning environments, reverberation can impair a listener's comprehension of speech due to poor speech intelligibility, especially when listeners are hard of hearing, expending a greater auditory workload to process speech resulting in reduced task performance.<sup>7-12</sup> In rooms where teleconferencing or audio equipment introduces amplified sound, spaces without adequate acoustical treatment are subject to higher functioning background noise levels, which may be heard in neighboring spaces.<sup>113-15</sup> Lack of reflected sound control in spaces for conferencing can impact the quality of speech intelligibility over teleconferencing equipment.

**Solutions :** Providing acoustical treatment that significantly reduces sound across human speech frequencies can greatly reduce unwanted buildup of reflected sound.<sup>1</sup> Ceilings typically provide the greatest area of coverage where highly absorptive materials can offer the best performance in reducing reverberation and controlling speech intelligibility and occupied noise levels.<sup>3,16,17</sup> Wall treatment can reduce reflections that impact performance of speech intelligibility with teleconferencing equipment.<sup>1</sup>

### PART 1 IMPLEMENT SOUND REDUCING SURFACES (MAX : 2 PT )

*For All Spaces except Dwelling Units:*

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

a. Acoustical furnishings meet the criteria shown in the following table:

| Room Type                           | Metric                                      | Tier 1<br>1 Point                               | Tier 2<br>2 Points                                                                                                                                                    |
|-------------------------------------|---------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open workspaces                     | Minimum NRC OR Alpha-w                      | 0.75 for at least 75% of available ceiling area | 0.90 for all available ceiling area <sup>1,18</sup>                                                                                                                   |
|                                     | Minimum furniture height and NRC OR Alpha-w | N/A                                             | Partial height barriers with a minimum height of 1.2 m above finished floor and a minimum NRC OR Alpha-w value of 0.70 between all opposing workstations <sup>1</sup> |
| Areas for conferencing and learning | Minimum NRC OR Alpha-w at ceilings          | 0.75 for at least 50% of available ceiling area | 0.90 for all available ceiling area                                                                                                                                   |
|                                     | Minimum NRC OR Alpha-w at walls             | 0.75 on at least 25% of one wall                | 0.80 on at least 25% of two perpendicular walls                                                                                                                       |
| Areas for dining                    | Minimum NRC OR Alpha-w at ceilings          | 0.75 on at least 50% of available ceiling area  | 0.90 for all available ceiling area                                                                                                                                   |

### REFERENCES

1. General Services Administration. Sound Matters: How to Achieve Acoustic Comfort in the Contemporary Office. 2011. [https://www.wbdg.org/FFC/GSA/gsa\\_soundmatters.pdf](https://www.wbdg.org/FFC/GSA/gsa_soundmatters.pdf).
2. Bradley. Speech Intelligibility Studies in Classrooms. J Acoust Soc Am. 1986.
3. Bradley JS. The acoustical design of conventional open plan offices. Can Acoust. 2003;27(3):23-31. doi:NRCC-46274 A
4. Venetjoki N, Kaarlela-Tuomaala A, Keskinen E, Hongisto V. The effect of speech and speech intelligibility on task performance. Ergonomics. 2006;49(11):1068-1091. doi:10.1080/00140130600679142
5. Jensen KL, Arens E, Zagreus L. Acoustical Quality in Office Workstations , As Assessed By Occupant Surveys. Indoor Air 10th Int Conf indoor air Qual Clim. 2005;2401-2405. doi:10.1111/j.1600-0668.2011.00745.x
6. Kim J, de Dear R. Workspace satisfaction: The privacy-communication trade-off in open-plan offices. J Environ Psychol. 2013;36:18-26. doi:10.1016/j.jenvp.2013.06.007
7. Klatte M, Hellbrück J, Seidel J, Leistner P. Effects of Classroom Acoustics on Performance and Well-Being in Elementary School Children: A Field Study. Environ Behav. 2010;42(5):659-692. doi:10.1177/0013916509336813
8. Bennetts L, Flynn M. Improving the classroom listening skills of children with Down syndrome by using sound-field amplification. Downs Syndr Res Pract. 2002;8(1):19-24.

9. Bradley JS. Speech intelligibility studies in classrooms. *J Acoust Soc Am*. 1986;80(3).
10. Cheesman MF, Jennings MB, Klinger L. Assessing communication accessibility in the university classroom: Towards a goal of universal hearing accessibility. *Work*. 2013;46(2):139-150.
11. Bradley JS. Speech intelligibility studies in classrooms *Speech intelligibility*. 2009;846(1986). doi:10.1121/1.393908
12. Peters LJ. Auditory Performance: A Model to Predict Task Performance as a Function of Auditory Workload: Overview. In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. ; 1991.
13. InfoComm International. *AV / IT Infrastructure Guidelines for Higher Education*.
14. Larsen JB, Vega A, Ribera JE. The Effect of Room Acoustics and Sound-Field Amplification on Word Recognition Performance in Young Adult Listeners in Suboptimal Listening Conditions. *Am J Audiol*. 2008;17(1):50-59.
15. Watson A, Sasse MA. Measuring Perceived Quality of Speech and Video in Multimedia Conferencing Applications. :1-6.
16. Pierrette M, Parizet E, Chevret P, Chatillon J. Noise effect on comfort in open-space offices: development of an assessment questionnaire. *Ergonomics*. 2015;58(1):96-106. doi:10.1080/00140139.2014.961972
17. Larm P, Keränen J, Helenius R, Hakala J, Hongisto V. Acoustics in open-plan offices -- A laboratory study. *Forum Acusticum*. 2005;(Figure 1):2021-2025.
18. Veitch JA, Bradley JS, Legault LM, Norcross SG, Svec JM. IRC-IR-846: Masking speech in open-plan offices with simulated ventilation noise: Noise level and spectral composition effects on acoustic satisfaction. 2002; (November):1-56.